# Who benefits most?

Sorted effects, step by step · A colon cancer trial

Valentin Klotzbücher · Department of Clinical Research

# Session overview

1. Is the effect the same for everybody?
2. Sorted effects, step by step: a colon cancer trial.
3. Can we trust the pattern?
4. Discussion.

# Is the effect the same for everybody?

01

A trial reports one average effect. Whom does it describe?

# One average, many patients

- A trial answers: does the treatment work **on average**?
- Patients differ, so the effect can differ too: large for some, small or none for others.
- “Why not … just estimate the distribution of effects?”
- Then we can ask: **who benefits most, and who least?**

Quote from *The Effect*, §22.3 (Huntington-Klein 2025).

# Same average, different stories

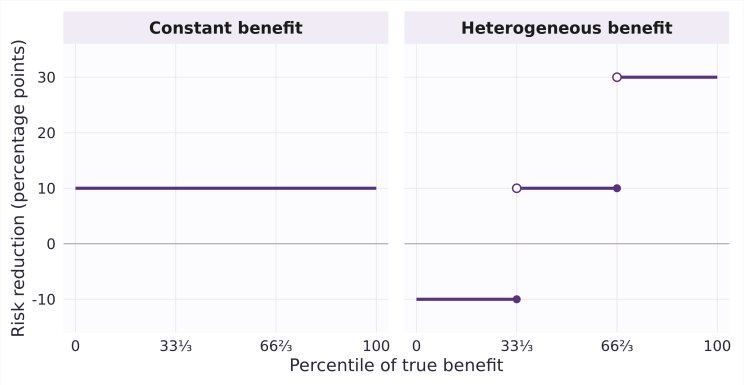
Three equally common kinds of patients, A, B and C. Without treatment, 40% of each die.

| Benefit (pp) | A | B | C |
|---|---|---|---|
| Everyone benefits equally | +10 | +10 | +10 |
| Benefit differs | -10 | +10 | +30 |

Both stories: average benefit **10 pp**. The average cannot tell them apart.

Constructed example with known risks, not data. Benefit = risk without treatment − risk with treatment.

# Sorting tells them apart

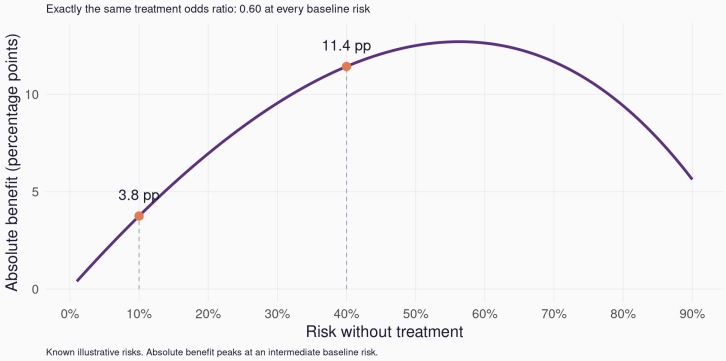



- Line patients up from least to most benefit.
- Equal benefit: a flat line at **10 pp**.
- Different benefit: a staircase from **−10 to +30 pp**.
- One third are worse off, despite a positive average.

In a trial we must estimate this curve. That is what sorted effects do.

Exact quantiles of the constructed example; no estimation.

# Even a constant odds ratio gives different benefits




Algebraic illustration with known risks.

- Same odds ratio (**0.60**) for every patient.
- **10%** baseline risk → **3.8 pp**.
- **40%** baseline risk → **11.4 pp**.
- A logistic model already lets benefits differ; interactions let them differ more.

# Sorted effects, step by step

02

A colon cancer trial: who benefits from chemotherapy?

# The trial

- 619 patients with resected **stage III colon cancer**, randomized to observation or **levamisole + 5-FU**.
- Outcome: **death within three years**.
- Died: **34.7%** with observation (109/314), **25.7%** with chemotherapy (78/304).
- On average, about **9 fewer deaths per 100** treated patients. The same for everybody?

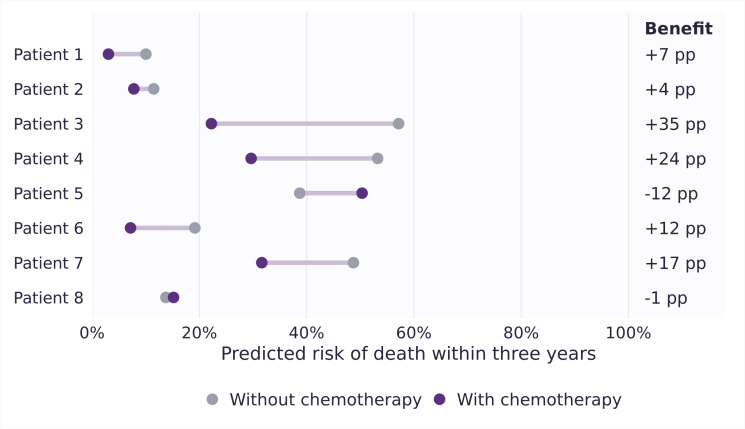
`survival::colon`, from the trial reports (Moertel et al. 1990, 1995). The third arm, levamisole alone, is not used. Analyses use the 593 patients with complete baseline data.

# Step 1 · Let the effect differ

- A logistic model for death within three years.
- Chemotherapy’s effect may change with **age, sex, number of positive nodes, tumour grade and tumour extent**.
- Bowel obstruction, tumour adherence to nearby organs and time from surgery to registration adjust the baseline risk.
- Randomization makes the comparison fair; the model describes how the effect varies.

Six treatment interactions, fixed before fitting; 19 parameters. Model code in backup.

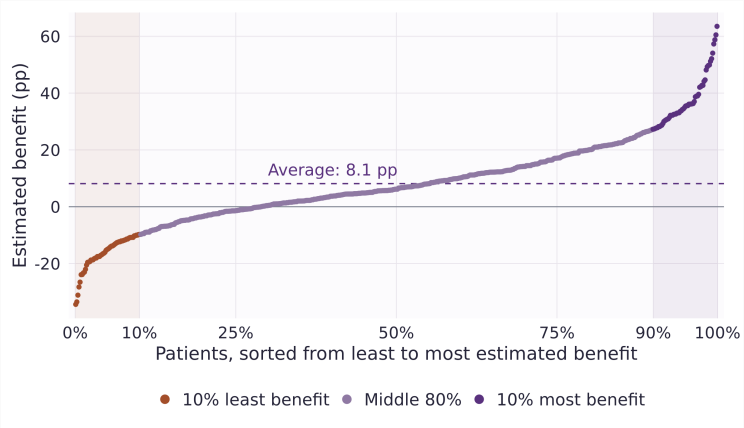
# Step 2 · Predict both futures for every patient




- For each patient: predicted risk **without** and **with** chemotherapy.
- The gap is the **estimated benefit**.
- It is an average for patients like them, not a personal effect.

Eight of 593 patients, chosen to span the range and shown unsorted. Point estimates from the Step 1 model.

# Step 3 · Sort them




- All 593 patients, from least to most estimated benefit.
- Average: **8.1 pp**. Middle 80%: **−10 to +27 pp**.
- Only **15%** of patients are within 3 pp of the average.
- The extremes are the least certain estimates.

Point estimates; uncertainty follows in chapter 03. In R: `SortedEffects::spe()` (call in backup).

# Step 4 · Who is at each end?

| Characteristic | 10% least benefit | 10% most benefit | Adjusted p |
|---|---|---|---|
| Women | 100% | 8% | .04 |
| Women under 55 | 92% | 0% | .04 |
| Mean age (years) | 43 | 53 | .87 |
| More than 4 positive nodes | 13% | 88% | .13 |
| Poor differentiation | 10% | 42% | .95 |

- Least benefit: younger women with few positive nodes. Most: mostly men with many nodes.

This describes who sits at each end. It is not the effect of chemotherapy in women.

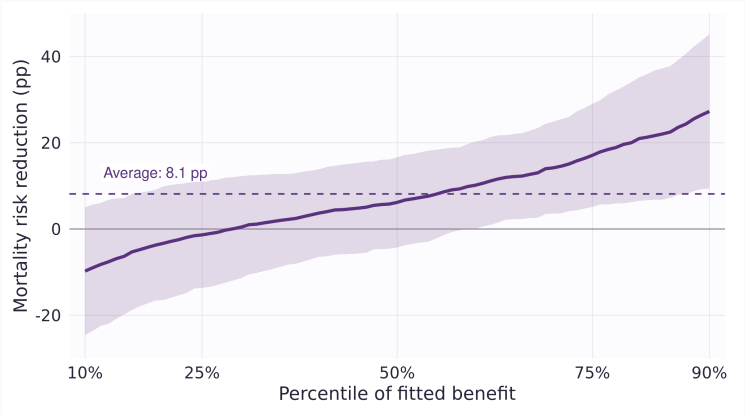
60 patients per tail. p-values are adjusted for comparing 12 traits. The package's own bootstrap broke down for these data, so they come from our checked re-run with stable starting values (backup).

# Can we trust the pattern?

03

Is the spread real, or noise from one model and one trial?

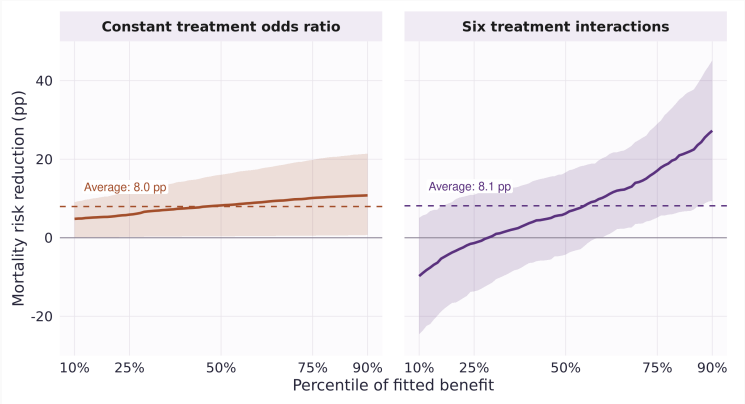
# How sure are we?




- Shaded: a 95% band for the **whole curve** at once.
- Near the bottom (10th percentile): **−9.8 pp**, band **−24.6 to +5.1**: we cannot tell whether these patients benefit.
- Average: **8.1 pp**, 95% CI **1.4 to 14.9**.

Band from the package's bootstrap (499 resamples), covering the 10th–90th percentiles. How it is built: backup.

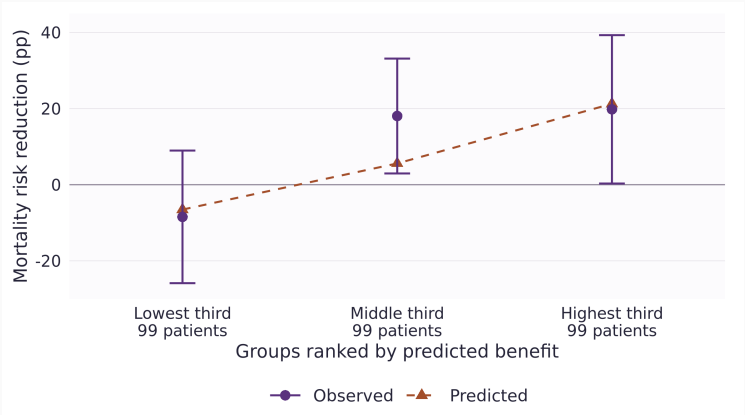
# You find what your model allows




95% bands for the whole curve, one per model.

- Constant odds ratio: benefit varies only with baseline risk, a narrow spread.
- Six interactions: a much wider spread.
- Same average (**8.0 vs 8.1 pp**); the six interactions improve the fit ($p = .014$).
- More spread is not proof: noise spreads too.

# Does it hold up in new patients?




- Fit on half the patients (296); rank the other half (297) blind to their outcomes.
- Higher-ranked patients did benefit more: slope **1.1** (0 = no signal, 1 = as predicted), p = .025, adjusted for checking two models.
- Lowest third: **−8.5 pp**, CI −25.9 to +9.0: again unclear.

One random split, balanced by treatment arm; three groups of 99; a 95% interval for each group.

# What could fool us?

- **Noise looks like spread:** 19 parameters for 179 deaths; 60 patients per tail.
- **Missing data:** keeping all 618 patients moves the p-value for the interactions from **.014 to .054**.
- **One split, one trial:** and colon was the third trial we tried, with no adjustment for that search.

A promising ranking to test in other trials, not a treatment rule.

# Discussion

04

What would make sorted effects useful in our own trials?

# What did we learn?

- The average (**8 pp**) hid a wide range of estimated benefits.
- Sorting turned a 19-parameter model into one curve and a table of who sits at each end.
- The pattern is uncertain and depends on the model; it needs other trials.

# Questions for discussion

- **Spread or noise?** 29% of patients get a negative estimated benefit. Would you shrink the interactions first?
- **Planning:** would you prespecify sorted effects in a trial's analysis plan?
- **Risk or interactions:** start from baseline risk, or let the effect depend on traits?
- **Scale:** absolute or relative benefit?

# Next time: causal forests

- **Here:** we chose which interactions to allow.
- **Causal forests:** let the data choose. A tree splits patients where the effect differs most, say under and over 65; a forest averages many trees.
- **Same caveat:** neither method creates causality. Both need a design, here randomization.

*The Effect*, §22.3.1 (Huntington-Klein 2025); generalized random forests (Athey, Tibshirani, and Wager 2019). No forest is fitted in this session.

# Reading and materials

- **Intuition:** *The Effect*, chapter 10 (treatment effects) and §22.3 (heterogeneous effects), free online.
- **Tutorial:** the SortedEffects R package (Chen et al. 2020).
- **Method:** the sorted effects paper (Chernozhukov, Fernández-Val, and Luo 2018).
- **Code, report and slides:** github.com/valentinklotzbuecher/jcod.

# Additional material

+

Code, bootstrap details, full tables and assumptions.

# Backup · The estimand

$$\Delta(x) = \mathsf{Pr}(Y = 1 \mid A = 0, X = x) - \mathsf{Pr}(Y = 1 \mid A = 1, X = x)$$

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)]$$

- $Y$: death within three years; $A = 1$: levamisole + 5-FU, $A = 0$: observation; $x$: the eight baseline characteristics in the model.
- With randomization, $\Delta(x) = E[Y(0) - Y(1) \mid X = x]$: an average over patients with profile $x$, not an individual effect.
- The distribution of $X$, here the 593 observed profiles, defines the population summarized.
- SPE(u), the sorted partial effect, is the u-th percentile of the effects; APE is their average.
- A quantile of effects differs from a difference of outcome quantiles.

# Backup · The model in R

```
1  fm <- outcome ~ treatment + splines::ns(age, df = 3) + female +
2    age_z:female + log_nodes + poor_diff + extensive +
3    obstruct + adhere + surg_late +
4    I((treatment == "chemotherapy") * age_z) +
5    I((treatment == "chemotherapy") * female) +
6    I((treatment == "chemotherapy") * age_z * female) +
7    I((treatment == "chemotherapy") * log_nodes) +
8    I((treatment == "chemotherapy") * poor_diff) +
9    I((treatment == "chemotherapy") * extensive)
10
11 fit <- glm(fm, data = dat, family = binomial())
```

**1–3:** common model · **4–9:** six treatment interactions · **11:** fit the model.

Comparison model: omit lines 4–9. `age_z = (age - 60) / 10`; `log_nodes = log1p(nodes)`.

# Backup · Predict both conditions in R

```r
control <- treated <- dat
control$treatment <- "observation"
treated$treatment <- "chemotherapy"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep every patient's baseline characteristics fixed.
- Change only treatment assignment.
- Subtract predicted **risks**, not log odds.

# Backup · Sorted effects with spe()

```r
spe_result <- SortedEffects::spe(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  us = seq(.10, .90, .01),
  b = 499,
  alpha = .05, bc = FALSE,
  seed = 420
)
```

**Fit the outcome model**
Our chosen formula and a logistic link.
**Set the contrast**
This ordering gives observation − chemotherapy.
**Choose the percentiles**
10th–90th, in one-percentile steps.
**Quantify uncertainty**
499 resamples; 95% band; no bias correction.

Uniform coverage concerns the chosen percentile grid and requires valid effect estimation and regularity conditions.

# Backup · How is the uniform band built?

1. Refit in each of **499** bootstrap draws and re-sort the fitted benefits at 81 percentiles, 10th–90th.
2. Scale each percentile by its bootstrap interquartile range ÷ 1.349.
3. Critical value: 95th percentile of the **largest** scaled deviation across all 81 percentiles.
4. Band: estimate ± critical value × scale; endpoints rearranged to stay monotone.

| Model | Uniform critical value | Pointwise |
|---|---|---|
| Six treatment interactions | 2.79 | 1.96 |
| Constant treatment odds ratio | 2.17 | 1.96 |

The package's own bootstrap (SortedEffects 1.7.0), which we reproduced independently; critical values read off the saved bands. Rearrangement: Chernozhukov, Fernández-Val, and Galichon (2009).

# Backup · Comparing the tails with ca()

```r
ca_result <- SortedEffects::ca(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  u = .10,
  interest = "moment", cl = "diff",
  t = colon_characteristics()$variable,
  b = 499, alpha = .05,
  bc = FALSE, seed = 420
)
```

**Model and contrast**
Same mortality model; benefit = observation − chemotherapy.
**Choose the tails**
Bottom and top 10%: 60 patients each here.
**Compare characteristics**
High − low means, across all twelve traits.
**Bootstrap inference**
499 resamples; SEs and p-values adjusted across the twelve traits.

`ca()` is classification analysis. For these data its own bootstrap breaks down; the p-values we show come from our checked re-run with stable starting values (next slide).

# Backup · Why the package's tail bootstrap failed here

- In each bootstrap resample, `ca()` refits the logistic model with weights scaled to sum to **20,000**.
- On these data, **141 of 499** refits did not converge, and **137** resamples gave undefined tail means.
- The function still printed normal-looking SEs and p-values. We do not use them.
- Our re-run starts each refit from stable values: **all 499** refits converge, and none are dropped.

Check every bootstrap refit: output that looks normal can still be wrong.

Same resamples, model, tail definitions and tail means; only the numerical starting point changed. Details in the colon report.

# Backup · Calculate tail means

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
traits <- colon_characteristics()$variable
lapply(tails, function(i)
  colMeans(dat[i, traits]))
```

- Select tails by fitted benefit, then summarize baseline characteristics.
- Both tails can contain patients from either randomized arm.

Descriptive calculation only. `R/colon-walkthrough.R` verifies these means against the saved classification table.

# Backup · Classification: demographics and nodes

| Characteristic | Low | High | Difference | SE | Joint p |
|---|---|---|---|---|---|
| Age (years) | 43.2 | 52.9 | +9.7 | 7.1 | 0.872 |
| Female (%) | 100.0 | 8.3 | -91.7 | 21.4 | 0.042 |
| Men aged 65+ (%) | 0.0 | 6.7 | +6.7 | 14.2 | 1.000 |
| Women under 55 (%) | 91.7 | 0.0 | -91.7 | 21.1 | 0.040 |
| Positive nodes | 2.6 | 8.4 | +5.8 | 2.2 | 0.297 |
| More than 4 nodes (%) | 13.3 | 88.3 | +75.0 | 22.1 | 0.132 |

Difference = high − low. Percentages differ in percentage points.

60 patients per tail. SEs and p-values from our checked re-run of the tail bootstrap (stable starting values, 499 resamples); p-values adjusted across all twelve traits, not just these six.

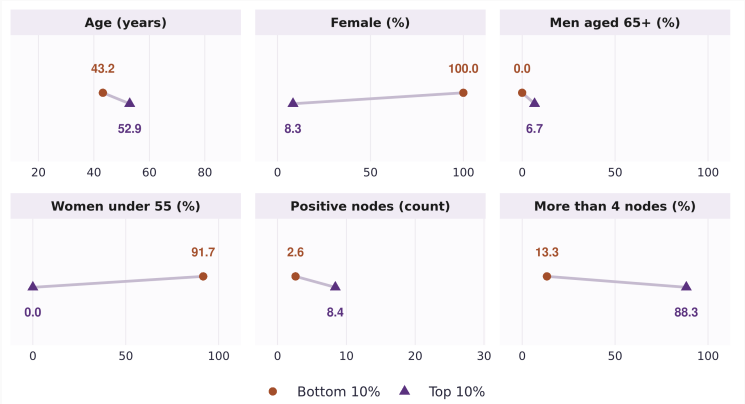
# Backup · Classification: clinical characteristics

| Characteristic | Low | High | Difference | SE | Joint p |
|---|---|---|---|---|---|
| Poor differentiation (%) | 10.0 | 41.7 | +31.7 | 27.7 | 0.948 |
| Serosa or beyond (%) | 98.3 | 85.0 | -13.3 | 17.4 | 0.992 |
| Bowel obstruction (%) | 35.0 | 20.0 | -15.0 | 9.6 | 0.762 |
| Perforation (%) | 3.3 | 3.3 | +0.0 | 3.7 | 1.000 |
| Tumour adherence (%) | 16.7 | 15.0 | -1.7 | 10.5 | 1.000 |
| Long surgery interval (%) | 33.3 | 25.0 | -8.3 | 10.8 | 0.990 |

Difference = high − low, in percentage points.

“Long surgery interval”: surgery-to-registration category. SEs and p-values from our checked re-run of the tail bootstrap, adjusted across all twelve traits.

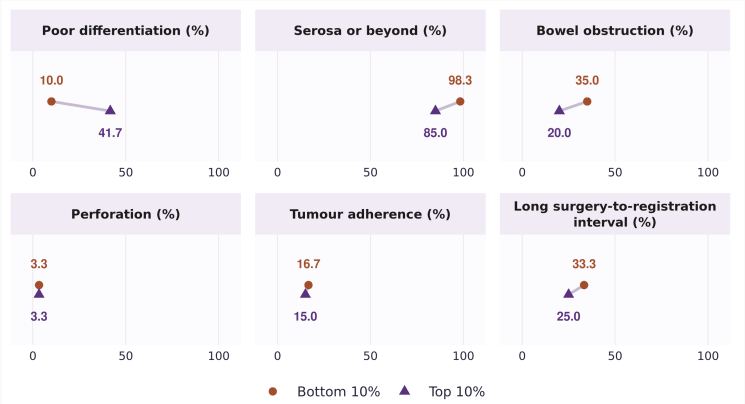
# Backup · Tail profiles: age, sex and nodes




- Low tail: **all women**, mean age **43**.
- High tail: **92% men**, mean age **53**.
- More than four nodes: **13% vs 88%**.
- Men aged 65+: **7% of the high tail**.

Descriptive means; 60 patients per tail. Connecting segments are not confidence intervals.

# Backup · Tail profiles: clinical characteristics




“Adherence” denotes tumour adherence to nearby organs.

- Poor differentiation: **10% vs 42%**.
- Bowel obstruction: **35% vs 20%**.
- Perforation: **3% in both tails**.
- Adjusted p-values: classification tables.

# Backup · What is resampled?

- `spe()`**:** refits the model in each resample, then sorts the predicted effects for the original 593 patients.
- `ca()`**:** refits the model in each resample, recomputes who falls in each tail, and recomputes the tail means.
- **Held-out check:** fits on one half only, freezes the ranking, then looks at outcomes in the other half.

# Backup · Cohort and sensitivity checks

| Analysis | Patients | Average benefit | Interaction p |
|---|---|---|---|
| Complete baseline and endpoint | 593 | 8.1 pp | .014 |
| Missing baseline values with indicators | 618 | 8.4 pp | .054 |
| Early-censored outcome assigned 0 | 594 | 8.0 pp | .017 |
| Early-censored outcome assigned 1 | 594 | 8.4 pp | .010 |

- The latter two retain complete baseline variables.
- Eight records disagree on the supplied vs count-derived four-node indicator; the report includes a coding check.

# Backup · When is the heterogeneity causal?

- **Identification, both:** randomization or no unmeasured confounding; overlap; consistency/no interference; appropriate handling of selection and missing data.
- **Regression → sorted effects and tail comparison:** reliable conditional-risk estimates; suitable convergence and regularity of the effect distribution.
- **Forests:** enough local data, regularity, suitable nuisance estimates and honest fitting.
- **Inference:** pointwise forest intervals do not automatically cover sorted effects or selected tails.

Both concern conditional average effects. Fitted spread alone does not establish true heterogeneity. (Chernozhukov, Fernández-Val, and Luo 2018; Athey, Tibshirani, and Wager 2019)

# Backup · Where do sorted effects fit?

| Approach | Question | Watch for |
|---|---|---|
| Subgroup forest plot | Does the effect differ by one trait? | One trait at a time; many tests |
| Risk modeling | Does absolute benefit change with baseline risk? | Misses modifiers unrelated to risk |
| Effect modeling | Which traits modify the effect? | Overfitting; needs validation |
| Sorted effects + classification | How are fitted effects distributed, and who is in the tails? | Only as good as the effect estimates |

Here, the constant-odds-ratio model is risk-based; the six interactions are effect modeling.

Risk and effect modeling as defined in the PATH statement (Kent et al. 2020).

# Backup · How a forest builds a comparison

For a patient profile, which treated and control patients are comparable?

**1 · Grow a tree**

Split baseline profiles to find groups with different treatment–control contrasts.

**2 · Grow many trees**

Different samples and candidate splits give different groups of similar patients.

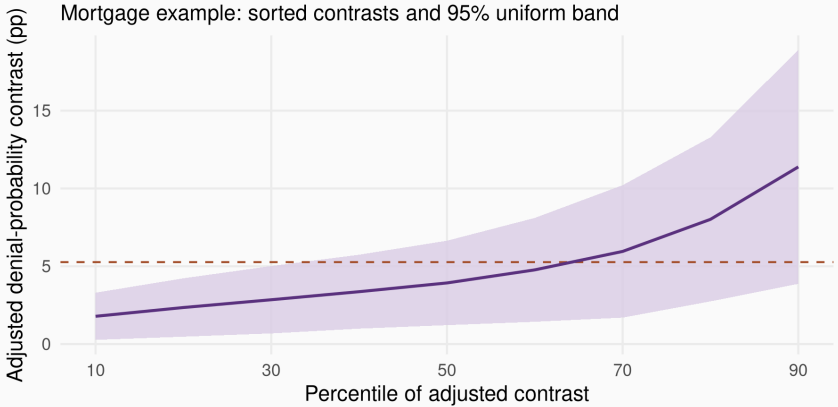
**3 · Combine the comparisons**

Patients repeatedly grouped with the target profile get more weight in its treatment comparison.

Many trees reduce dependence on any one noisy split.

“Honest” fitting uses separate observations to choose splits and estimate effects. Randomized-trial intuition; no forest was fitted here. (Athey, Tibshirani, and Wager 2019)

# Backup · Original published example




Tutorial benchmark: adjusted observational disparity (Munnell et al. 1996; Chen et al. 2020).

# References I


Athey, Susan, Julie Tibshirani, and Stefan Wager, “Generalized random forests,” *The Annals of Statistics*, 47 (2019), 1148–1178.

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, “SortedEffects: Sorted causal effects in R,” *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Alfred Galichon, “Improving point and interval estimators of monotone functions by rearrangement,” *Biometrika*, 96 (2009), 559–575.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, “The sorted effects method: Discovering heterogeneous effects beyond their averages,” *Econometrica*, 86 (2018), 1911–1938.

Huntington-Klein, Nick, “The Effect: An introduction to research design and causality,” (2025).

Kent, David M., Jessica K. Paulus, David van Klaveren, and others, “The predictive approaches to treatment effect heterogeneity (PATH) statement,” *Annals of Internal Medicine*, 172 (2020), 35–45.

Moertel, Charles G., Thomas R. Fleming, John S. Macdonald, and others, “Levamisole and fluorouracil for adjuvant therapy of resected colon carcinoma,” *New England Journal of Medicine*, 322 (1990), 352–358.

# References II

——, “Fluorouracil plus levamisole as effective adjuvant therapy after resection of stage III colon carcinoma: A final report,” *Annals of Internal Medicine*, 122 (1995), 321–326.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, “Mortgage lending in Boston: Interpreting HMDA data,” *The American Economic Review*, 86 (1996), 25–53.

van Klaveren, David, Ewout W. Steyerberg, Patrick W. Serruys, and David M. Kent, “The proposed ‘concordance-statistic for benefit’ provided a useful metric when modeling heterogeneous treatment effects,” *Journal of Clinical Epidemiology*, 94 (2018), 59–68.